MILITARY SPECIFICATION SHEET

STARTER, AIR TURBINE, ACCESSORY DRIVE COMBINATION,
AIRCRAFT ENGINE, MODEL ATS/AD-2

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the starter described herein shall Consist of this document and the latest issue of Specification MIL-S-22518(AS)

REQUIREMENTS

Applicable paragraphs
of MTL-S-22518A(AS)

3.8.1	Envelope: Figure 1
3.7	Wt, Max (Lb): 55
4.5.19	Aircraft Model: A-7
4.5.20	Engine Model: TF41
3.8.2.2	Accessory Drive: Engine: Figure 1 Pad "B": Figure 1
3.5.5 3.9.1	Rotation Viewed from Pad "B" End: Pad "A": Clockwise Pad "B": Counterclockwise
3.3.2	Ambient Temperature Operating Range (°F): Starter: -65 to +160 Accessory Drive: -65 to +250
3.3.2	Exposure Temperature Range (°F): -100 to +275
3.3.2.2 3.5.7 4.5.1.1 4.5.7	Control Valve: Opening Rate, Max: 22 psig per second pressure rate for first 0.5 second; full open in 3 to 5 seconds Closing Rate, Max: 0.5 second Navy Model: ATSCV- 1

FSC 2995

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REQUIREMENTS (Continued)

Applicable paragraphs
of MIL-S-22518A(AS)

3.5.1 Rated Conditions:

Starter (Pad "A"):
Drive Speed (RPM): 1200
Drive Torque, Min (Lb-Ft): 290
Airflow, Max (Lb per Min): 110
Inlet Air Total Pressure (In. Hg A): 100
Inlet Air Total Temperature (°F): 345
Exhaust Gas Discharge Pressure (In. Hg A): 29.9

Accessory Drive (Pad "B"):
Continuous: 20 HP at 4300 RPM
Intermittent: 40 HP at 4300 RPM for 5 minutes
Maximum: 55 HP at a Pad "A" speed of 6800 RPM for period specified in 4.5.7

NOTE : At an input speed of 7200 RPM at pad "A" drive, pad "B" drive shall rotate at 4320 ±10 RPM.

3.5.1 cutout RPM: 3200 plus 200 minus 0

3.5.2 Maximum Temperature Limit: The air turbine shall be
3.5.8 capable of operating at a maximum inlet air total temperature limit of 650°F at rated inlet air total pressure.

3.5.8 Altitude: The ATS/AD shall not be damaged when subjected
4.5.17 to 75,000 ft altitude in the overrunning mode of operation. The ATS/AD shall be capable of developing rated drive speed and torque at an 8000 ft altitude in the starter mode of operation. Operating as an accessory drive the ATS/AD shall complete 20 cycles of the test schedule of Figure 2.

3.5.9 Attitude: The ATS/AD shall be capable of meeting
4.5.16 the operational requirements when the ATS/AD longitudinal axis, through the pad "A" shaft, is in any position within 10 degrees of the horizontal position. Inverted flight test time is 15 minutes.

3.5.10 Overrunning (Qualification): The ATS/AD shall be
4.5.10 overrun at a 7250 RPM pad "A" drive speed for a total of 20 hours. Testing need not be continuous but test periods shall not be less than 5 hours.

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REQUIREMENTS (Continued)

Applicable paragraphs
of MIL-S-22518A(AS)

3.5.10
4.5.10 Overrunning (Acceptance): The ATS/AD shall be overrun at a pad "A" speed of 8000 RPM and subjected to a pad "B" load of 20 HP for 5 minutes.

4.5.4 Initial Calibration:

Pad "A" Drive

Torque (Lb-Ft)	Speed RPM	Speed Condition
420 Min, 465 Max	0	Stall
290 Min ---	1200	Rated
50 Min ---	3400	cutout

Airflow, Max (Lb per Min)	speed RPM	Speed Condition
110	C	Stall

4.5.5 No-Load Operation (Qualification):
Time (Second): 60
Pad "A" Drive Speed (RPM), Min: 3600

4.5.5.3 Burst Test:
Inlet Air Total Pressure (PSIA): 75
Inlet Air Total Temperature: 445 ±10°F

4.5.6 No-Load Operation (Acceptance): Pad "A" Drive Speed (RPM), Min: 3600

3.5.5
4.5.7 Endurance Test:

Accessory Drive: The ATS/AD shall be driven for 960 hours at 7200 RPM with a continuous load of 20 HP applied at pad "B." A load of 40 HP shall be applied at pad "B" for 5 minutes of every hour during the above 960-hour endurance test. This phase of endurance testing need not be continuous; however, test periods shall not be less than 5 hours duration. This endurance testing may be combined with the tests for the starter in this paragraph in an attempt to simulate service conditions.

Accessory Drive High Torque Test: At a pad "A" drive speed of 6800 RPM the ATS/AD shall be cycled for 15 minutes at 20 HP and 15 minutes to 55 HP for a total of 2 hours.

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REQUIREMENTS (Continued)

Applicable paragraphs
of MIL-S-22518A(AS)

Starter: No. of cycles: 1200. Each cycle shall consist of three consecutive phases as follows:

Test Stand Rotor Polar Moment of Inertia	Phase A - Acceleration		Phase B - At Following Conditions			Phase C - Acceleration	
	Drive Speed* Min	Time Max	Drive Speed*	Load Torque Min	Time	Drive Speed* Min	Time
Lb-Ft ^{2}	RPM	Sec	RPM	Lb-Ft	Sec	RPM	Sec
690	1200	8	1200	290	10	2350	23

*Pad "A"

NOTE: The above values do not allow for flywheel windage losses, test stand drag, or inertia of the ATS/AD. In order to determine compliance with this specification, the test equipment shall be calibrated and specific test results corrected accordingly. If the acceleration specified for Phase C is not developed because of test stand drag torque, the test stand inertia load may be reduced or external power may be applied to assist acceleration to cutout speed. Time for end of phase C to cutout shall not exceed 15 seconds.

4.5.9 Rated Torque Calibration:
Pad "A" Drive Speed (RPM), Min: 1200
Airflow, Max (Lb per Min): 110

3.5.10
4.5.10 Overrunning: The ATS/AD shall be overrun at a pad "A" drive speed of 7250 RPM for a total of 20 hours. Testing need not be continuous, but test periods shall not be less than 5 hours.

4.5.11.1 Low Temperature Operation:
Starter: The inlet air pressure and temperature and the ATS/AD stabilized temperature for the low temperature test shall be as follows:

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REQUIREMENTS (Continued)

Applicable paragraphs
of MIL-S-22518A(AS)

Inlet Air Total Pressure In. Hg Abs ±0.5	Inlet Air Total Temperature °F ±10	ATS/AD Temperature °F (Maximum)
140	260	-65

After the ATS/AD temperature has been stabilized for a minimum of 8 hours at -65°F ambient temperature, three consecutive cycles at a -65°F ambient temperature shall be run. Each cycle shall consist of the following phases:

Phase A: The starter shall accelerate the flywheel having a polar moment of inertia of 690 lb-ft^3 for a period of 8 seconds from rest.

Phase B: After a period of 8 seconds from rest, a torque load shall be applied to maintain the unit starter output RPM constant at ±100 RPM for a period of 10 seconds.

Phase C: At the end of phase B the torque load shall be removed and the ATS/AD allowed to accelerate the flywheel to cutout speed or for an additional period of 12 seconds, whichever, occurs first. Upon completion of the 12-second period or phase C if cutout speed has not been obtained, the ATS/AD shall be motored by an external source to a speed that till cause the automatic shutoff controls to operate.

Accessory Drive: After a minimum of 8-hour soak period at -65°F, the ATS/AD shall be operated at -65°F ambient temperature at 7200 RPM input at pad "A" for a period of 10 hours with a 20 HP continuous load applied at pad "B." A 40 HP load shall be applied for the last 5 minutes of the 10-hour cycle.

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REQUIREMENTS (Continued)

Applicable paragraphs
of MIL-S-22518A(AS)

4.5.11.2 High Temperature Operation:
Starter: The inlet air pressure and temperature and the stabilized ATS/AD temperatures for the high temperature test shall be as follows:

Inlet Air Total. Pressure Tn. Hg Abs ±0.5	Inlet Air Total Temperature °F ±10	ATS/AD Temperature °F (Minimum)
90	445	160

After the ATS/AD temperature has been stabilized for a minimum of 8 hours at +160°F, three consecutive cycles at a +160°F ambient air temperature shall be run. Each cycle shall be as specified for phases A, B, and C of 4.5.11.1.

Accessory Drive: After a minimum 8-hour soak period at an ambient air temperature of +250°F, the ATS/AD shall be operated with ambient air at 250°F minimum at 7200 RPM speed at pad "A" for a period of 30 minutes. A 20 HP continuous load shall be applied at pad "B," except for 2.5 minutes a load of 40 HP shall be applied. The 30-minute cycle of operation shall be repeated until 10 hours of operating time are accumulated.

4.5.12 Consecutive Cycle Operation: No. of cycles: 5

4.5.14 Vibration: All vibration testing shall be conducted with the specified accessory or a dummy, having a weight of 17.5 pounds and a center of gravity 4 inches from the mounting face, mounted on the accessory pad (pad "B"), Mounting shall be dynamically similar to that encountered in service use. Vibration tests shall be conducted in accordance with MIL-STD-810, Method 514, Procedure 1, Curve F (Parts 1, 2, and 3). In addition, the ATS/AD shall be capable of withstanding a loading of 10g in any direction without structural damage.

3.5.6
4.5.18 Pad "A" Drive Shear Section Strength: Torque (Lb-Ft): 1000 +50

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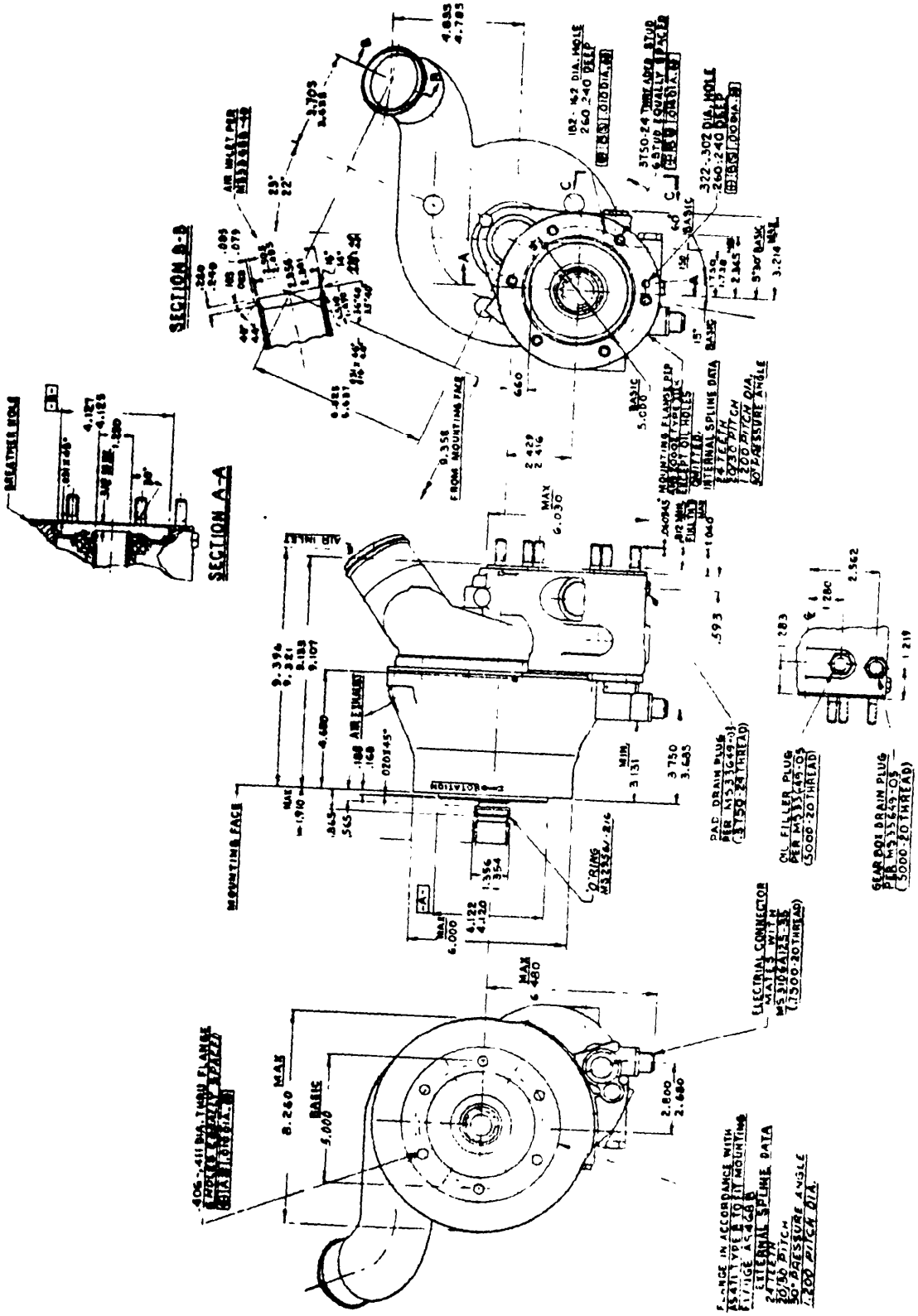


FIGURE 1. Starter, Air Turbine, Model ATS/AD-2.

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TEST SCHEDULE FOR ATS/AD-1 OPERATING AS AN ACCESSORY DRIVE

ALTITUDE (FT)	TIME (MIN)	PAD "B" DRIVE SPEED (RPM)	HYDRAULIC PUMP DISCHARGE PRESSURE (PSI)	HYDRAULIC PUMP FLOW (GPM)	ATS/AD AMBIENT AIR TEMPERATURE °F	CONDITION
Sea Level	7.5	2980	2950	13	200	Idle
Sea Level	7.5	2980	2950	0	200	Idle
SL to 45,000	2.5	4230	2950	13	200	Military
SL to 45,000	2.5	4230	2950	0	200	Military
45,000	1	4230	2950	18	200	Military
45,000	1	4230	2950	6	200	Military
45,000	5	4230	2950	0	200	Military
45,000	1	4230	2950	18	200	Military
45,000	1	4230	2950	6	200	Military
45,000	6	4230	2950	0	200	Military
45,000	1	4100	2950	18	170	*Normal
45,000	1	4100	2950	6	170	*Normal
45,000	5	4100	2950	0	170	*Normal
45 to 70,000	1	4260	2950	6	200	Afterburner
45 to 70,000	4	4260	2950	0	200	Afterburner
70,000	1	4260	2950	6	200	Afterburner
70,000	4	4260	2950	0	200	Afterburner
70 to 45,000	1	4230	2950	18	200	Military
70 to 45,000	1	4230	2950	0	200	Military
70 to 45,000	1	4230	2950	18	200	Military
70 to 45,000	1	4230	2950	0	200	Military
70 to 45,000	1	4230	2950	18	200	Military
45,000	1	4230	2950	18	170	**Normal
45 to 5000	1	4230	2950	0	170	Normal
45 to 5000	1	2980	2950	13	200	Idle
45 to 5000	4	2980	2950	0	200	Idle
45 to 5000	1	2980	2950	13	200	Idle
45 to 5000	4	2980	2950	0	200	Idle
5000	1	3680	2950	13	200	80%
5000	4	3680	2950	0	200	80%
5000	1	3680	2950	13	200	80%
5000	4	3680	2950	0	200	80%
5000	1	4230	2950	18	200	Military
5000	1	4230	2950	13	225	Military
5000	1	4230	2950	6	225	Military
5000	2	4230	2950	0	225	Military
5000	1	3600	2950	18	200	80%
5000	3	3600	2950	13	200	80%
5000	1	3600	2950	0	200	80%

Note: *Repeat for total of four cycles with zero load of last cycle for 7 minutes.

**Continue this run for 30 minutes with l-minute cycles at 18 GPM and momentarily back to zero flow between each l-minute loading.

FIGURE 2. Test schedule for ATS/AD-2.

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